

Graph Classes and Width Parameters in Lax

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This guide points to definitions in existing Lax submissions. Each marked passage opens the original concept and its Lean encoding beside the PDF. The source identifiers are also clickable in the PDF itself. The chapters are a reading order, not new definitions or claims of equivalence between formalizations. This submission has no concepts or proofs of its own.

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1 Graph classes and closure

Graph classes. The sparsity lectures use families of finite simple graphs on canonical vertex sets; see [lax-199508](#).

Isomorphism-closed graph classes. A separate submission develops class operations in an isomorphism-invariant setting; see [lax-871432](#).

Closure properties. This concept covers induced-subgraph, minor, contraction, summand, and union closure. Induced-subgraph closure is the hereditary-class property; see [lax-871432](#).

Graph parameters. A common signature and functional equivalence of parameters appear in [lax-153141](#).

2 Sparse graph classes

Nowhere dense classes. The source uses excluded shallow clique minors; see [lax-199508](#).

Uniform quasi-wideness. The distance-independent-set viewpoint is in [lax-199508](#).

Weakly sparse classes. Exclusion of a complete bipartite subgraph is formalized in [lax-710763](#).

Monadically dependent classes. The transduction-based class property appears in [lax-710763](#).

Planar graphs. A straight-line drawing certificate is used in [lax-68](#).

Outerplanar graphs. The corresponding drawing-based concept is [lax-68](#).

Series-parallel graphs. Two-terminal series and parallel composition is treated in [lax-68](#).

3 Hereditary and forbidden-pattern classes

Cographs. This submission characterizes the class by twin-width zero; see [lax-214022](#).

Induced-minor exclusion. The containment relation itself is in [lax-762056](#).

Excluded induced grid minors. A class specified by forbidding an induced square-grid minor is represented in [lax-762056](#).

Chromatic versus clique number. The χ -bounded class property is formalized in [lax-9](#).

4 Width and deletion parameters

Treewidth. Tree decompositions and their width are defined in [lax-228581](#).

Twin-width. The partition-sequence formulation is in [lax-228581](#).

Mixed minor number. This ordered-adjacency-matrix parameter is in [lax-153141](#).

Merge-width. Merge sequences and radius-dependent width appear in [lax-9](#).

Cliquewidth. The k -expression formulation is in [lax-271696](#).

Feedback vertex number. Deletion of vertices to an acyclic graph is measured in [lax-379983](#).

Feedback edge number. The edge-deletion counterpart is in [lax-379983](#).

5 Sparsity measures and vertex orders

Generalized coloring numbers. Weak and strong reachability through short paths are treated in [lax-199508](#).

Admissibility. The path-family parameter is in [lax-199508](#).

Shallow-minor density. A depth-dependent edge-density measure is in [lax-199508](#).

Almost-linear neighborhood complexity. Neighborhood traces on vertex sets are formalized in [lax-199508](#).

Linear neighborhood complexity. A separate formulation appears in [lax-9](#).

6 Containment and example families

Graph minors. The branch-set formulation is in [lax-68](#).

Topological minors. The internally disjoint path formulation is in [lax-68](#).

Grids and walls. Canonical sparse examples appear in [lax-68](#).

Trees. The tree class is recorded in [lax-68](#).

Paths. The finite path graphs are recorded in [lax-68](#).